

Structured Synthetic Supervision and Metric Pitfalls in Ultra-Low-Resource Tangut Short-Text Translation

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Abstract

Tangut translation is an extreme low-resource problem: in the current repository snapshot, only 491 real Tangut-Chinese pairs are available, with 400/41/50 train-dev-test examples. The task is also structurally unusual because many targets are short, title-like Chinese strings rather than ordinary modern sentences. We evaluate three families of methods in this setting: inference-only prompting, synthetic-data supervised fine-tuning (SFT), and direct preference optimization (DPO). Structured multitask SFT is the strongest pure supervised baseline, but the strongest current learned system comes from stricter gap-filtered DPO on top of that base. With only 498 preference pairs whose reward gap is at least 0.4, the sigmoid variant reaches chrF++ 30.01, keeps 5/50 exact matches, preserves 22/27 title suffixes, and ties for the best overall score (2.22/5) under a new reference-aware judge. We further show that several apparently reasonable reference-free metrics are misaligned with the task: the gold human reference itself receives only moderate dictionary coverage and extremely poor perplexity under a modern-Chinese language model, despite achieving perfect chrF++ and perfect reference-aware judge scores. At the same time, preference optimization is not uniformly helpful: legacy DPO on a weaker base still degrades reference-aligned performance, and looser gap filtering yields adequacy gains only with substantial length or contamination costs. These findings suggest a two-stage practical lesson for ultra-low-resource Tangut short-text translation: explicit structural supervision is the right starting point, preference optimization can help only when pair quality is aggressively controlled, and evaluation must be anchored to reference-aware metrics rather than generic fluency proxies.

1 Introduction

Machine translation for historical and ultra-low-resource languages is difficult for two separate reasons. First, parallel data is often scarce or nearly absent. Second, the outputs that matter to scholars are not always long, natural modern sentences: they may be titles, glosses, liturgical fragments, or compact bibliographic strings. Recent work on ancient-language NLP and historical-text translation has made this setting more visible, but it has also shown that data scarcity and evaluation instability are especially severe outside modern benchmark conditions [1, 2, 3]. Tangut is a canonical example. The script is historically important, but usable parallel Tangut-Chinese resources remain tiny. In the current repository snapshot, we only have 491 real Tangut-Chinese pairs.

This small-data regime creates a practical question: when full supervised translation is impossible, what kind of signal is most helpful? A natural first option is dictionary-augmented prompting, in the spirit of lexicon-guided translation. A second option is to synthesize pseudo-parallel training data from dictionary knowledge and monolingual Chinese. A third option is to add preference optimization on top of an SFT base [5, 4, 7]. The repository already contains experiments spanning all three options, which makes it possible to write a focused empirical paper from the existing evidence.

Our main finding is now two-stage. Structured synthetic supervision is still the strongest pure SFT recipe: the multitask variant that exposes intermediate alignment structure during training remains the best supervised baseline. But the strongest overall learned systems appear only after preference optimization is moved onto that stronger base and the preference pairs are filtered much more aggressively. In particular, the strict gap-filtered DPO sigmoid variant reaches the highest learned-system chrF++ in the repository (30.01), ties for the best reference-aware overall score (2.22/5), keeps 5 exact matches, and stays close to reference title length.

The repository also reveals two important failure modes. First, reference-free metrics can be badly misleading. The human reference itself is penalized by lexical coverage and perplexity, which means those metrics cannot serve as primary model-selection criteria for this task. Second, preference optimization is highly conditional. Legacy DPO on a weaker base underperforms badly, and even multitask-base DPO remains unstable when low-gap preference pairs are retained. The positive result therefore depends on both the base model and the pair-quality filter, not on DPO in the abstract.

The current draft therefore makes three contributions:

1. an empirical comparison of prompting, synthetic SFT, and DPO for Tangut short-text translation;
2. a positive method result showing that structured multitask SFT is the right base and that strict gap-filtered DPO can surpass it on raw overlap while matching the best reference-aware quality;
3. a reference-aware evaluation and alignment analysis showing why reference-free metrics and low-gap preference data can fail.

The rest of the paper situates the task against related work, describes the experimental setting and the three method families, and then separates the positive result from the metric and alignment failure analysis that makes the result interpretable.

2 Related Work

Ancient and historical language processing. Recent surveys argue that ancient-language NLP differs from mainstream NLP not just because data are scarce, but because scripts, editorial conventions, and annotation practices are unstable across corpora [1]. Shared tasks on ancient-language translation have likewise emphasized that even when translation is the end goal, the available supervision is often tiny and domain-specific [2]. Work on historical document modernization faces related sparsity and spelling-shift issues, but often targets orthographic or lexical modernization rather than cross-script title reconstruction [3]. Our Tangut setting is even narrower: only 491 real pairs are available, and the evaluation set is dominated by compact title-like strings.

Synthetic supervision and lexical grounding. Low-resource machine translation often relies on synthetic parallel data built from monolingual corpora, with back-translation as the canonical example [4]. A separate line of work injects lexical knowledge more directly, either by folding bilingual lexicons into the model or by constraining decoding to respect required terminology [5, 6]. Our prompting and SFT baselines sit at the intersection of these ideas: they exploit dictionary knowledge and synthetic supervision, but the central question is whether stronger gains come from input purity, approximate semantic preservation, or explicit alignment structure.

Preference optimization. Direct Preference Optimization (DPO) has become a standard offline objective for aligning generation models without a separate reward-model-plus-RL loop [7]. More recent work has pushed the same idea into machine translation, for example by using query-specific rewards for NMT or by constructing MT preferences from word alignment and MBR decoding [10, 11]. Another line of work studies robustness to noisy preference labels through modified DPO objectives or importance-style reweighting [8, 9]. Our results connect these threads in a harsher regime: the supervision is tiny, the outputs are short title-like strings, and the chosen/rejected signal is built from partially misaligned automatic metrics. In that setting, DPO can help, but only after base-model choice and pair filtering are controlled much more aggressively.

Evaluation for non-standard translation outputs. chrF++ is widely used in machine translation because character-level overlap is often more stable than token-level overlap under morphological or orthographic variation [12]. Learned evaluation and quality-estimation frameworks such as COMET and OpenKiwi push beyond surface overlap by predicting translation quality from richer signals [13, 14]. Our findings add a complementary caution: in short historical title translation, apparently reasonable reference-free metrics can still mis-rank the gold reference itself. This is why the current paper treats reference-aware evidence as primary and reference-free metrics as diagnostic.

3 Task Setting

The real Tangut data in this repository contains 491 pairs. The current split is 400 training examples, 41 development examples, and 50 test examples. Unlike ordinary machine-translation benchmarks, the targets are short and often title-like. Many examples resemble book titles, sutra titles, or compact scholastic expressions. This matters because title reconstruction rewards precise lexical choices and compact formatting, while a modern Chinese fluency model may incorrectly treat those outputs as unnatural.

Because the real dataset is tiny, the system uses synthetic data to train translation models. For each SFT branch, 50,000 synthetic Tangut-Chinese pairs are produced from monolingual classical Chinese and dictionary knowledge, then combined with the 400 real training examples to form a 54,000-example training set. All learned models use the same instruction-tuned 7B backbone.

We evaluate with two classes of metrics. The repository’s existing metrics are lexical coverage, perplexity, chrF++, and a dictionary-conditioned LLM judge. chrF++ is a reasonable overlap anchor for this setting because character-based scoring is relatively stable under orthographic variation [12]. Learned MT evaluation and quality-estimation systems can capture richer signals than surface overlap [13, 14], but in this repository the available reference-free signals are only partially aligned with the task. For that reason, this draft does not treat all metrics equally: chrF++, exact match, contamination, suffix preservation, and the new reference-aware judge are primary evidence, while lexical coverage, perplexity, and the legacy reference-free judge are reported as diagnostics.

4 Methods

4.1 Inference-only baselines

The first family contains no weight updates. **baseline1** directly prompts the base model with the Tangut string. **baseline2** augments the prompt with dictionary glosses. **baseline2.1_cot** further restructures the prompt into a three-step process intended to encourage alignment, literal drafting,

and rewriting. This family is closest in spirit to lexicon-augmented or terminology-aware MT, except that the lexicon is injected through prompting rather than through decoding constraints or model-side features [5, 6].

These methods are important because they show what can be achieved from symbol injection alone. In the current results, dictionary prompting is substantially better than zero-shot generation, but it often expands compact titles into explanatory paraphrases rather than reconstructing the reference form.

4.2 Synthetic SFT baselines

The second family uses synthetic training data.

- **baseline3** uses mixed Tangut-Chinese inputs.
- **baseline3_1.unk** enforces pure Tangut inputs by replacing dictionary misses with UNK.
- **baseline3_3.semantic** also enforces pure inputs, but replaces misses through vector-space semantic projection instead of UNK.
- **baseline3_2.multitask** introduces a structured target containing explicit alignment scaffolding.

The central design question is whether the model benefits more from input purity, approximate semantic preservation, or explicit alignment structure. At a high level, this family plays the role that synthetic parallel data often plays in low-resource MT: it converts monolingual or weakly aligned resources into supervised signals that a translation model can absorb [4]. The current evidence favors explicit alignment structure.

4.3 Preference optimization

The final family applies DPO to an SFT base using reward-defined chosen/rejected pairs [7]. In the current repository, the reward combines lexical coverage with a penalty on log perplexity. We study this family in two stages because the repository now contains both a failure case and a repaired variant.

The legacy pipeline first selects the best SFT base among the mixed, UNK, and semantic variants using chrF++, then constructs preference pairs from that base. In the current snapshot, that selector chooses the UNK model, yielding the legacy checkpoint `final.v2`. This run is still important because it exposes the failure mode directly: if the base model is weaker and the reward is partly misaligned, DPO can degrade title reconstruction rather than improve it.

The follow-up DPO runs start instead from the strongest supervised base, the multitask SFT model. After auditing the preference data, we remove invalid rows and keep only pairs whose reward gap exceeds a threshold, testing both a looser filter (≥ 0.2 , 1,996 pairs) and a stricter filter (≥ 0.4 , 498 pairs). For each filtered set we train two variants: standard sigmoid DPO and a noise-aware alternative that combines TRL’s robust DPO loss with WPO-style reweighting for off-policy preferences [8, 9]. This turns the DPO family into a controlled diagnosis of three factors at once: base-model choice, pair-quality filtering, and loss robustness.

Method	Lex $\uparrow$	PPL $\downarrow$	chrF++ $\uparrow$	Sem $\uparrow$	Flu $\uparrow$
Zero-shot	0.1165	20.21	0.19	1.00	1.56
Dict prompt	0.8187	4474.44	6.13	2.30	2.40
Dict+CoT	0.8723	18119.16	7.59	2.30	1.88
Mixed-SFT	0.3192	298684.91	17.85	1.42	1.88
UNK-SFT	0.3514	11031.18	22.11	1.66	1.96
MT-SFT	0.4839	3322.93	25.99	1.70	2.26
Sem-SFT	0.3723	17064.38	17.03	1.46	1.84
Legacy DPO	0.4224	314577.16	19.39	1.52	1.76
MT-DPO g0.2 sig	0.5694	303438.00	20.93	1.88	1.86
MT-DPO g0.2 rob	0.5249	238892.42	27.67	1.88	1.88
MT-DPO g0.4 sig	0.4833	9573.71	30.01	1.84	2.04
MT-DPO g0.4 rob	0.4547	20012.69	27.52	1.74	2.08
Human ref.	0.5733	1646139.34	100.00	2.24	2.34

Table 1: Repository-level metrics on the 50-example Tangut test set. Here **g0.2** and **g0.4** denote minimum preference-gap thresholds of 0.2 and 0.4. Bold marks the best learned-system value in each column, excluding prompt-only baselines and the human reference. The strongest pure SFT baseline is MT-SFT, but stricter gap-0.4 filtering changes the DPO story again: MT-DPO g0.4 sig gives the highest learned-system chrF++ and later ties for the best reference-aware overall score.

5 Main Results

Table 1 summarizes the repository metrics. The central pattern is no longer a simple “SFT good, DPO bad” split. Zero-shot generation still fails, dictionary prompting is still strong on coverage-like diagnostics, and structured multitask SFT remains the strongest pure supervised baseline. But once DPO is rerun on the multitask base with reward-gap filtering, the story changes again: strict filtering does not just rescue DPO qualitatively, it becomes the strongest overall learned configuration in the repository.

The most important raw-metric comparison now spans `baseline3.2_multitask`, the looser gap-0.2 DPO variants, and the stricter gap-0.4 follow-ups. The strongest new result is the strict gap-0.4 sigmoid DPO variant. Tightening the reward-gap threshold from 0.2 to 0.4 cuts the preference set from 1,996 pairs to 498, but it raises learned-system chrF++ to 30.01, preserves 5 exact matches, and brings the average output length back near the reference ratio (1.12). The strict robust+weighted variant is slightly weaker on chrF++ (27.52) but is still much cleaner than its gap-0.2 counterpart. By contrast, the looser gap-0.2 sigmoid model already showed reference-aware adequacy gains, yet it overgenerated badly (length ratio 1.74), while the looser robust+weighted model preserved suffixes most aggressively but contaminated 7 outputs.

These disagreements are precisely why the paper should not rely on a single automatic metric. Raw chrF++, contamination, and the later reference-aware judge each reward different parts of the short-title problem. In the current Tangut setting, the more defensible claim is now stronger but also more specific: structured SFT is the best starting point, and high-gap multitask-base DPO can improve on it. Loose filtering favors some adequacy gains at the cost of unstable surface behavior, while strict filtering yields the best overall raw/clean trade-off.

Method	Exact Match	Contam. Rate	Suffix Preserve	Len. Ratio
Dict prompt	0/50	0.00	3/27	1.83
UNK-SFT	3/50	0.06	17/27	1.05
MT-SFT	5/50	0.02	19/27	1.10
Legacy DPO	2/50	0.18	18/27	1.88
MT-DPO g0.2 sig	5/50	0.00	19/27	1.74
MT-DPO g0.2 rob	4/50	0.14	23/27	1.58
MT-DPO g0.4 sig	5/50	0.04	22/27	1.12
MT-DPO g0.4 rob	5/50	0.04	22/27	1.26
Human ref.	50/50	0.00	27/27	1.00

Table 2: Reference-aligned diagnostics derived from stored predictions. “Contam.” denotes mixed-script or artifact contamination. In the last two columns, the table reports suffix-preservation count first and length ratio second. Bold marks the best learned-system value in each column, excluding prompt-only baselines and the human reference. The new gap-0.4 variants keep most of the suffix gains of multitask-base DPO while restoring near-reference length and sharply reducing contamination relative to their gap-0.2 counterparts.

6 Analysis

6.1 Reference-aligned diagnostics

Repository-level metrics alone do not fully explain the new ranking. Additional diagnostics from stored predictions make the trade-off much clearer. The multitask model remains the cleanest pure SFT system, with 5 exact matches, only 1 contaminated output, and a near-reference output length ratio (1.10). The looser gap-0.2 sigmoid DPO follow-up keeps the same 5 exact matches and drops contamination to 0/50, but stretches titles to 1.74 times the reference length. The looser robust+weighted follow-up instead preserves title suffixes most aggressively (23/27), but pays for that with 7 contaminated outputs.

The stricter gap-0.4 follow-ups change the surface behavior substantially. The strict sigmoid model keeps 5 exact matches, raises title-suffix preservation to 22/27, and restores the length ratio to 1.12 while reaching the highest learned-system chrF++ in the repository (30.01). The strict robust+weighted model also keeps 5 exact matches and 22/27 suffix preservation, but with only 2 contaminated outputs instead of 7. In other words, stricter pair filtering does not merely preserve the earlier DPO gains; it regularizes them into something much closer to the short-title target distribution.

By contrast, the dictionary-prompting baseline often produces outputs that are too long and too explanatory. Its average output is about 1.83 times the reference length. This helps lexical coverage but hurts exact title reconstruction.

6.2 Reference-aware judging changes the DPO story

The legacy LLM judge in the repository is reference-free, so we reran evaluation with a stricter judge that sees the Tangut input, the gold reference, and the candidate output simultaneously. This rerun serves two purposes. First, it checks whether the human reference is correctly ranked when the judge has the information it should need. Second, it tests whether the raw-metric ranking survives once the judge can penalize explanatory overgeneration and content drift directly.

Table 3 changes the paper narrative again. The human reference still receives perfect 5.0 scores on all dimensions, so the judge is well anchored. But among non-reference systems, the

Method	Ref. Agr.	Src. Faith.	Title Style	Overall
UNK-SFT	1.78	1.84	4.36	1.82
MT-SFT	1.92	1.98	4.18	1.98
Legacy DPO	1.64	1.72	3.80	1.68
MT-DPO g0.2 sig	2.08	2.24	4.14	2.22
MT-DPO g0.2 rob	2.10	2.16	3.78	2.12
MT-DPO g0.4 sig	2.12	2.26	4.60	2.22
MT-DPO g0.4 rob	2.10	2.20	4.70	2.22
Human ref.	5.00	5.00	5.00	5.00

Table 3: Reference-aware judge scores on a 1–5 scale. The judge sees the Tangut input, the gold reference, and the candidate output simultaneously. Bold marks the best learned-system value in each column, excluding the human reference. The best overall score is shared by three multitask-base DPO variants, but the stricter gap-0.4 runs win the tie-break dimensions: MT-DPO g0.4 sig leads on reference agreement and source faithfulness, while MT-DPO g0.4 rob leads on title-style fitness.

best overall score is now shared by three multitask-base DPO variants: the loose sigmoid run, the strict sigmoid run, and the strict robust+weighted run all reach 2.22. Within that tie, the strict gap-0.4 runs are stronger on title reconstruction: the strict sigmoid model reaches the highest reference-agreement score (2.12) and the highest source-faithfulness score (2.26), while the strict robust+weighted model reaches the highest title-style score (4.70). The older `final_v2` checkpoint remains worse than `baseline3.2_multitask`, so the positive result is specific to the stronger base and higher-quality pairs, not to DPO in the abstract.

The subset breakdown makes the new trade-off sharper. On the 27 reference titles containing common title suffixes, both strict gap-0.4 variants reach mean overall judge score 2.41, clearly above `baseline3.2_multitask` (2.15) and above the looser gap-0.2 variants (2.15 and 2.19). Just as importantly, they do so without the earlier surface penalties: the strict sigmoid run stays near reference length (1.12) and the strict robust run drives contamination back to 0.0% on this subset. On the 23 non-suffix items, the looser `final_gap02_multitask_sigmoid` remains strongest at 2.30, while both strict gap-0.4 variants score 2.00, still above the multitask SFT baseline (1.78). At the example level, `final_gap04_multitask_robustwpo` beats `baseline3.2_multitask` on the judge’s overall score for 17 items, loses on 4, and ties on 29; the strict sigmoid run wins 15, loses 4, and ties 31. This is strong evidence that DPO can help content alignment in this setting, but also that the reward-gap threshold controls which subset benefits most.

6.3 Human-reference anchor

One of the strongest pieces of evidence in the repository is the `human_reference` experiment. The gold reference receives perfect chrF++ and perfect reference-aware judge scores, as expected, but only moderate lexical coverage and catastrophic perplexity. This reveals a core evaluation issue: the current lexical coverage metric measures overlap with dictionary-derived semantics rather than agreement with the gold title, while the perplexity model penalizes compact title style and historical transliterations.

The implication is methodological, not cosmetic. If we were to select models purely by lexical coverage or perplexity, we would risk preferring systems that paraphrase, overgenerate, or stop poorly instead of systems that best reconstruct the true title. The new multitask-base DPO follow-ups make this especially clear: the loose gap-0.2 sigmoid run reaches top-tier reference-

Audit item	Value
Total preference pairs	3867
Duplicate chosen/rejected pairs	3
Non-finite reward entries	3
Mean reward gap	0.2396
Chosen closer to gold proxy	71.3%
Chosen closer on gap (0.05, 0.10]	62.3%
Chosen closer on gap (0.20, 0.40]	75.4%
Chosen closer on gap ≥ 0.40	81.5%
Pairs kept with gap ≥ 0.20	1996
Pairs kept with gap ≥ 0.30	1010
Pairs kept with gap ≥ 0.40	498

Table 4: Lightweight audit of the current DPO pair construction pipeline. Higher reward gaps correlate with cleaner preference labels under a simple gold-proxy check, which helps explain why strict filtering stabilizes the later multitask-base DPO runs.

aware quality while still overgenerating, whereas the strict gap-0.4 sigmoid run reaches the same 2.22 overall judge score while also delivering the best learned-system chrF++ and near-reference output length. Therefore, in the paper, lexical coverage and perplexity should be reported as supporting diagnostics, not as the primary decision criteria.

6.4 Legacy DPO fails for different reasons than multitask-base DPO

The old negative result is not a random bad run. The preference data itself is imperfect. The repository contains 3,867 preference pairs, including a small number of duplicate or invalid entries. More importantly, a quick audit against the original synthetic targets shows that the reward-chosen candidate is closer to gold only about 71% of the time under a simple similarity proxy, and that this quality rises monotonically with reward gap: 62.3% for gaps in $[0.05, 0.10]$, 66.3% for $(0.10, 0.20]$, 75.4% for $(0.20, 0.40]$, and 81.5% for gaps ≥ 0.40 . In other words, the high-gap filter is not a cosmetic hyperparameter. It is directly isolating cleaner preference labels.

The legacy selector also excludes the strongest multitask model from the DPO-base search. That turns out to matter. Once we switch to the multitask base and keep only pairs with reward gap ≥ 0.2 , DPO stops being a blanket negative result and begins to beat the multitask SFT baseline under the reference-aware judge. Tightening the filter further to gap ≥ 0.4 makes the mechanism much clearer. Relative to the loose gap-0.2 sigmoid model, the strict sigmoid model raises chrF++ from 20.93 to 30.01 and shrinks the length ratio from 1.74 to 1.12 while preserving the same 2.22 overall judge score. Relative to the loose robust+weighted model, the strict robust+weighted model reduces contamination from 7/50 to 2/50 and keeps the overall judge score at 2.22. Even a deterministic short-title normalization pass raises `final_gap02_multitask_robustwpo` only to chrF++ 29.06, still below the raw strict-sigmoid result. The reward is therefore not useless, but low-gap pairs inject exactly the kind of surface noise that a short-title system can least afford.

7 Discussion and Limitations

The paper should be explicit about scope. The current dataset supports Tangut short-text translation, especially title-like strings, not general open-domain Tangut sentence translation. The new subset analysis strengthens that boundary in a slightly different way than before: the largest gains

from strict gap-filtered DPO are concentrated on reference titles with canonical suffix patterns, while the looser gap-0.2 sigmoid variant remains stronger on the non-suffix subset. That scope is still meaningful: many historically important Tangut materials appear precisely in compact catalog, title, or scholastic forms, and low-resource methods for such data are valuable.

The paper should also be honest about absolute difficulty. Even the strongest learned system reaches only 2.22/5 under the fixed-key reference-aware judge, so the task is far from solved. The result is therefore not that Tangut title translation is easy, but that carefully structured supervision and carefully constrained preference optimization each help different parts of the problem.

Two limitations matter most for future work. First, the real dataset is extremely small, so the paper cannot claim stable behavior beyond the current narrow domain. Second, the new DPO result is still subset- and metric-sensitive: the strict gap-0.4 sigmoid variant gives the best raw/clean compromise, the strict robust+weighted variant gives the strongest title-style score, and the looser gap-0.2 sigmoid variant remains strongest on non-suffix adequacy. The most useful next experiments are therefore not “more DPO” in the abstract, but pair-selection curricula, title-aware stopping/decoding control, and training objectives that reward both adequacy and clean title form directly.

8 Conclusion

The current repository already supports a coherent paper, but the final message is no longer “SFT good, DPO bad.” Structured multitask synthetic supervision remains the strongest pure SFT baseline. Pure-input training is helpful, but noisy semantic projection does not yet outperform the simpler UNK strategy. On top of that stronger multitask base, however, stricter gap-filtered DPO now gives the strongest learned systems: `final_gap04_multitask_sigmoid` reaches the highest learned-system chrF++ (30.01) and ties for the best reference-aware overall score (2.22), while `final_gap04_multitask_robustwpo` reaches the same overall score with the strongest title-style fitness. The legacy `final_v2` checkpoint remains a genuine negative control, so the positive result is conditional on both the base model and the pair filter. Several reference-free metrics also remain misaligned enough to mis-rank the human reference. Together, these findings argue for a two-stage practical lesson: in ultra-low-resource Tangut short-text translation, explicit structural supervision is the right base recipe, aggressively filtered preference optimization can improve on it, and evaluation must stay anchored to reference-aware evidence.

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